10708
Probabilistic Graphical Models: Spring 2026

Andrej Risteski

Machine Learning Department

Lecture 13:
Fundamentals of
variational methods.
Inner and outer approximations

Algorithmic approaches to inference

When faced with a difficult to calculate probabilistic quantity (partition function, difficult posterior), there are two families of approaches:

MARKOV CHAIN MONTE CARLO

- Random walk w/ equilibrium distribution the one we are trying to sample from.
- **Pros**: Eventually, you will get samples from the intended distribution.
- **Cons**: Might need to run the algorithm very long; v. difficult to diagnose if the Markov Chain has converged.

VARIATIONAL METHODS

- Based on solving an optimization problem.
- **Pros:** Convergence is v. fast (lots of packages with v. efficient implementations)
- **Cons:** The optimization problems we are trying to solve are typically non-convex; unclear how good the solution we've converged to.

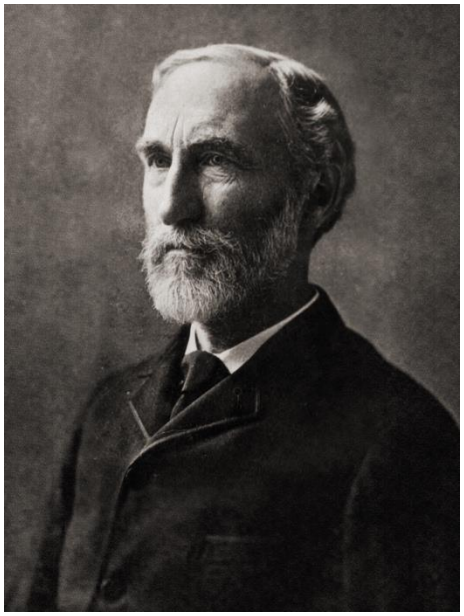
Variational methods for partition functions

Gibbs variational principle: Let $p(x) = \frac{1}{Z}\exp\big(E(x)\big)$ be a distribution over a domain $\mathcal{X}$. Then, Z is the solution to the following optimization problem:

$$\log\ Z = \max_{q:\ \text{distribution over } \mathcal{X}} H(q) + \mathbb{E}_{x\sim q}[E(x)]$$

Find the distribution that has both high entropy, and high expected energy value

$$H(q) \coloneqq -\sum_{x\in\mathcal{X}} q(x)\log q(x)$$



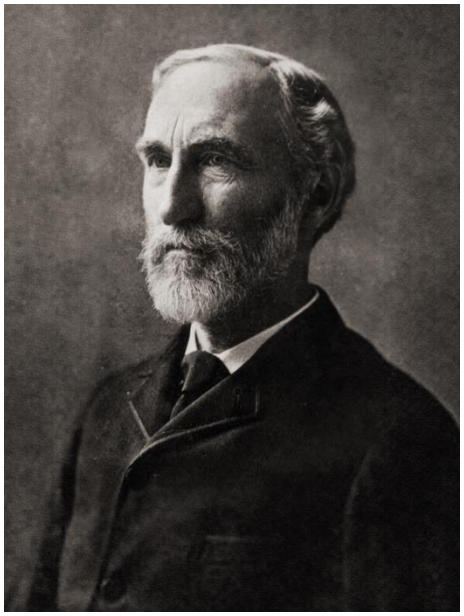
Variational methods for partition functions

Gibbs variational principle: Let $p(x) = \frac{1}{Z}\exp\big(E(x)\big)$ be a distribution over a domain $\mathcal{X}$. Then, Z is the solution to the following optimization problem:

$$\log\ Z = \max_{q\text{: distribution over } \mathcal{X}} H(q) + \mathbb{E}_{x\sim q}[E(x)]$$

Compare to MAP problem:

$$x^* = \mathrm{argmax}_x\ E(x)$$



Variational methods for partition functions

Gibbs variational principle: Let $p(x) = \frac{1}{Z}\exp\big(E(x)\big)$ be a distribution over a domain $\mathcal{X}$. Then, Z is the solution to the following optimization problem:

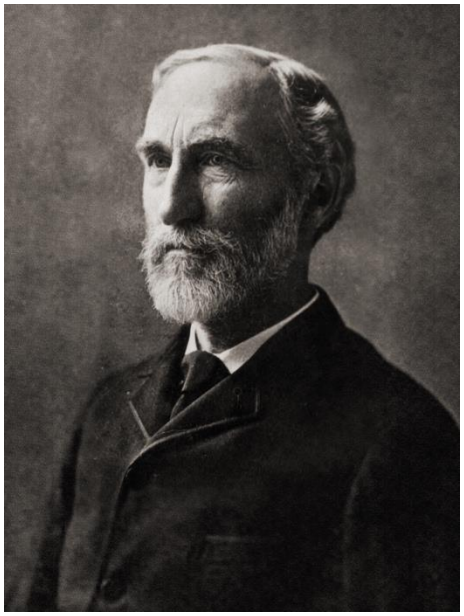
$$\log Z = \max_{q:\ \text{distribution over } \mathcal{X}} H(q) + \mathbb{E}_{x\sim q}[E(x)]$$

Proof: $0 \leq KL(q \,||\, p) = \mathbb{E}_q \log q - \mathbb{E}_q \log p$

$$= -H(q) - \mathbb{E}_{x\sim q}[E(x)] + \log Z$$

$$H(q) + \mathbb{E}_{x\sim q}[E(x)] = \log Z - KL(q||p) \leq \log Z$$

Hence, $\max_{q:\ \text{distribution over } \mathcal{X}} H(q) + \mathbb{E}_{x\sim q}[E(x)] \leq \log Z$



Variational methods for partition functions

Gibbs variational principle: Let $p(x) = \frac{1}{Z}\exp\big(E(x)\big)$ be a distribution over a domain $\mathcal{X}$. Then, Z is the solution to the following optimization problem:

$$\log Z = \max_{q:\ \text{distribution over } \mathcal{X}} H(q) + \mathbb{E}_{x\sim q}[E(x)]$$

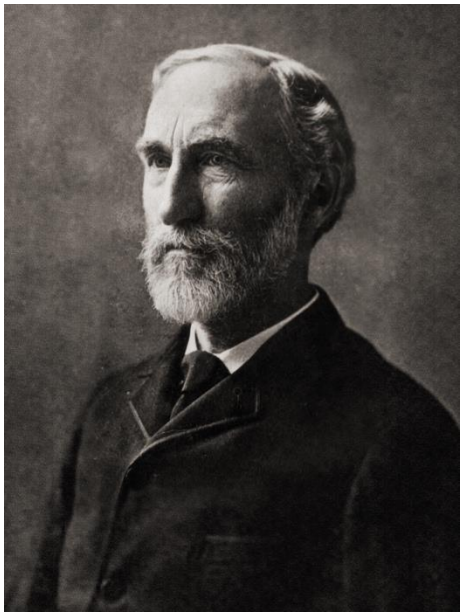
Proof: $0 \leq KL(q \,||\, p) = \mathbb{E}_q \log q - \mathbb{E}_q \log p$

$$= -H(q) - \mathbb{E}_{x\sim q}[E(x)] + \log Z$$

$$H(q) + \mathbb{E}_{x\sim q}[E(x)] = \log Z - KL(q||p) \leq \log Z$$

Equality is attained if p = q: $KL(q \,||\, p)$=0, so

$$H(q) + \mathbb{E}_{x\sim q}[E(x)] = \log Z$$



Variational methods for partition functions

Gibbs variational principle: Let $p(x) = \frac{1}{Z}\exp\big(E(x)\big)$ be a distribution over a domain $\mathcal{X}$. Then, Z is the solution to the following optimization problem:

$$\log Z = \max_{q:\ \text{distribution over } \mathcal{X}} H(q) + \mathbb{E}_{x\sim q}[E(x)]$$

Hence, we've reduced calculating partition function to an optimization problem!

But, there is a **serious issue**: how do we solve an optimization over the set of distributions over $\mathcal{X}$?

Even if $\mathcal{X}$ is a really simple domain, e.g. $\mathcal{X} = \{\pm 1\}^n$, the trivial way to solve the problem would involve introducing a variable $q(x), \forall x \in \{\pm 1\}^n$:
there are 2^n of them.

In fact, you can't be clever –
this can be #P hard even for Ising models!

Variational methods for partition functions

Gibbs variational principle: Let $p(x) = \frac{1}{Z}\exp\big(E(x)\big)$ be a distribution over a domain $\mathcal{X}$. Then, Z is the solution to the following optimization problem:

$$\log Z = \max_{q:\ \text{distribution over } \mathcal{X}} H(q) + \mathbb{E}_{x\sim q}[E(x)]$$

Hence, we've reduced calculating partition function to an optimization problem!

But, there is a **serious issue**: how do we solve an optimization over the set of distributions over $\mathcal{X}$?

Even if $\mathcal{X}$ is a really simple domain, e.g. $\mathcal{X} = \{\pm 1\}^n$, the trivial way to solve the problem would involve introducing a variable $q(x), \forall x \in \{\pm 1\}^n$:
there are 2^n of them.

We have to somehow solve a "simpler problem".
Two approaches to do so.

The physics approach: inner relaxation

Gibbs variational principle: Let $p(x) = \frac{1}{Z}\exp\big(E(x)\big)$ be a distribution over a domain $\mathcal{X}$. Then, Z is the solution to the following optimization problem:

$$\log Z = \max_{q:\ \text{distribution over } \mathcal{X}} H(q) + \mathbb{E}_{x\sim q}[E(x)]$$

What can we do to try to approximate this expression?

Inspiration from physics: solve a *simpler* optimization problem over a *restricted class* of distributions we can explicitly parametrize.

$$\log Z \approx \max_{q\in\mathcal{Q}} H(q) + \mathbb{E}_{x\sim q}[E(x)]$$

Example: naïve mean-field approximation: $\mathcal{Q}$ consists of product distribs

Consider again $\mathcal{X} = \{\pm 1\}^n$. A *product distribution* depends on n parameters only: since $\mathrm{q}(x) = \Pi_i q_i(x_i)$, for each $i \in [n]$, we only need to specify $q_i(x_i = 1)$.

The physics approach: inner relaxation

Gibbs variational principle: Let $p(x) = \frac{1}{Z}\exp\big(E(x)\big)$ be a distribution over a domain $\mathcal{X}$. Then, Z is the solution to the following optimization problem:

$$\log Z = \max_{q\text{: distribution over } \mathcal{X}} H(q) + \mathbb{E}_{x\sim q}[E(x)]$$

Inspiration from physics: solve a *simpler* optimization problem over a *restricted class* of distributions we can explicitly parametrize.

This is called an "*inner approximation*": it yields an approximation for $\log Z$ that is *less* than the true value (since we are maximizing over a smaller set than the original problem).

The maximization is generally not concave: so, optimization can get trapped in bad minima, saddle points, etc.

The physics approach: inner relaxation

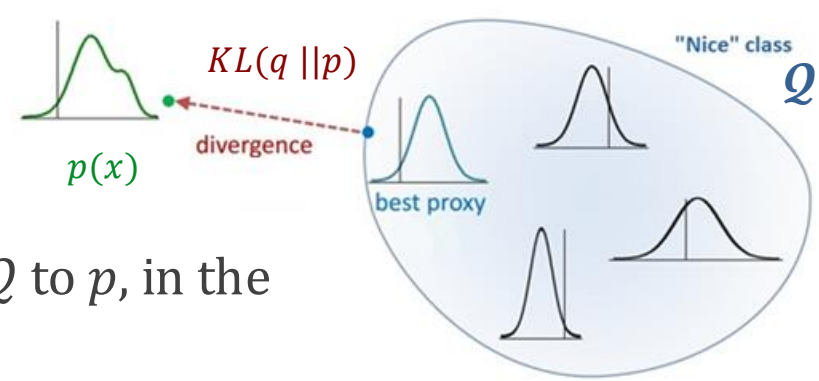
Gibbs variational principle: Let $p(x) = \frac{1}{Z}\exp\big(E(x)\big)$ be a distribution over a domain $\mathcal{X}$. Then:

$$KL(q \,||\, p) = -\big(H(q) + \mathbb{E}_{x\sim q}[E(x)]\big) + \log Z$$

Inspiration from physics: solve a *simpler* optimization problem over a *restricted class* of distributions we can explicitly parametrize.

Since $KL(q \,||\, p) = -\big(H(q) + \mathbb{E}_{x\sim q}[E(x)]\big) + \log Z$, we have:

$$\operatorname*{argmax}_{q\in\mathcal{Q}} H(q) + \mathbb{E}_{x\sim q}[E(x)]$$
$$= \operatorname{argmin}_{q\in\mathcal{Q}} KL(q \,||\, p)$$



We are finding the closest distribution $q \in \mathcal{Q}$ to p, in the KL divergence sense.

Solving the mean-field relaxation via coordinate ascent (CAVI)

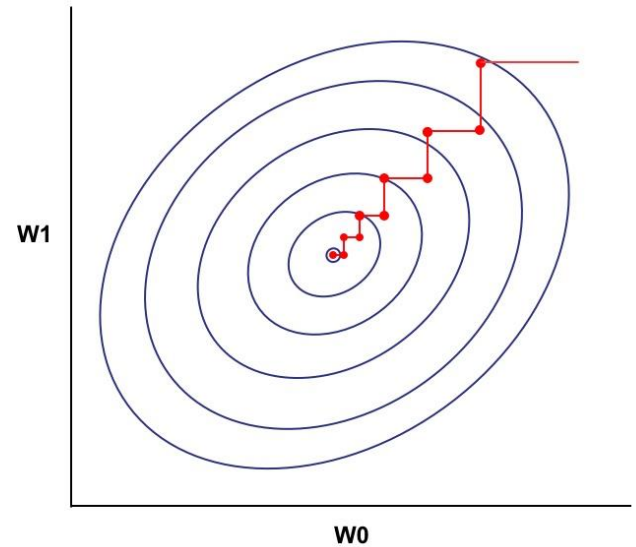
How do we solve a mean field relaxation?

A popular way to do so is <u>coordinate ascent:</u>

$$\underset{q \in \mathcal{Q}}{\operatorname{argmax}}\, H(q) + \mathbb{E}_{x \sim q}[E(x)]$$

$$= \underset{q \in \mathcal{Q}}{\operatorname{argmin}}\, \mathbb{E}_q \log q\,(x) - \mathbb{E}_q\, E(x)$$

$$= \underset{q \in \mathcal{Q}}{\operatorname{argmin}} \sum_i \mathbb{E}_{q_i} \log q_i\,(x_i) - \mathbb{E}_{q_i}\left[\mathbb{E}_{q_{-i}(x_{-i})} E(x)\right]$$



Consider holding q_j fixed for all $j \neq i$ and just updating q_i.

Dropping all terms not depending on q_i we are solving:

$$\underset{q_i}{\operatorname{argmin}}\, \mathbb{E}_{q_i} \log q_i\,(x_i) - \mathbb{E}_{q_i}\left[\mathbb{E}_{q_{-i}(x_{-i})} E(x)\right]$$

Solving the mean-field relaxation via coordinate ascent (CAVI)

$$\underset{q_i}{\operatorname{argmin}}\ \mathbb{E}_{q_i} \log q_i\,(x_i) - \mathbb{E}_{q_i}\left[\mathbb{E}_{q_{-i}(x_{-i})} E(x)\right]$$

What is the optimal q_i (among all univariate distributions)?

Let $Z_i := \int_{x_i} \exp\left(\mathbb{E}_{q_{-i}(x_{-i})} E(x)\right)$. Let $\widehat{p_i}(x_i) := \dfrac{\exp\left(\mathbb{E}_{q_{-i}(x_{-i})} E(x)\right)}{Z_i}$

$$\mathbb{E}_{q_i} \log q_i\,(x_i) - \mathbb{E}_{q_i}\left[\mathbb{E}_{q_{-i}(x_{-i})} E(x)\right]$$

$$= \mathbb{E}_{q_i} \log q_i\,(x_i) - \mathbb{E}_{q_i}\left[\log\ \widehat{p_i}(x_i)\right] - \log Z_i$$

$$= KL(q_i || \widehat{p_i}) - \log Z_i$$

The optimal q_i is $q_i = \widehat{p_i}$. *Can be efficiently calculated very often*.

Example: coordinate ascent updates for a pairwise UGM

Let's consider applying this strategy to $p(x) \propto \exp\left(\sum_{i\sim j} \phi_{ij}(x_i, x_j)\right)$

Then, $E(x) = \sum_{i\sim j} \phi_{ij}(x_i, x_j)$. Let's assume $x_i \in \{+1, -1\}$.

The updates involve setting $q_i(X_i = x_i) := \frac{\exp\left(\mathbb{E}_{q_{-i}(x_{-i})} E(x)\right)}{Z_i}$

We can calculate, for $x_i \in \{1, -1\}$, the value of $\mathbb{E}_{q_{-i}(x_{-i})} E(x)$

$$\mathbb{E}_{q_{-i}(x_{-i})} E(x) = \mathbb{E}_{q_{-i}(x_{-i})}\left(\sum_{j\in N(i)} \phi_{ij}(x_i, x_j) + \sum_{k,l:k\neq i, l\neq i} \phi_{ij}(x_i, x_j)\right)$$

$$= \mathbb{E}_{q_{-i}(x_{-i})}\left(\sum_{j\in N(i)} \phi_{ij}(x_i, x_j)\right) + C(x_{-i})$$

Example: coordinate ascent updates for a pairwise UGM

Let's consider applying this strategy to $p(x) \propto \exp\left(\sum_{i \sim j} \phi_{ij}(x_i, x_j)\right)$

Then, $E(x) = \sum_{i \sim j} \phi_{ij}(x_i, x_j)$. Let's assume $x_i \in \{+1, -1\}$.

The updates involve setting $q_i(X_i = x_i) := \frac{\exp\left(\mathbb{E}_{q_{-i}(x_{-i})} E(x)\right)}{Z_i}$

We can calculate, for $x_i \in \{1, -1\}$, the value of $\mathbb{E}_{q_{-i}(x_{-i})} E(x)$

$$\mathbb{E}_{q_{-i}(x_{-i})} E(x) = \mathbb{E}_{q_{-i}(x_{-i})}\left(\sum_{j \in N(i)} \phi_{ij}(x_i, x_j)\right) + C(x_{-i})$$

Hence, $q_i(X_i = x_i) \propto \exp\left(\mathbb{E}_{q_{i,N(i)}(x_{i,N(i)})}\left(\sum_{j \in N(i)} \phi_{ij}(x_i, x_j)\right)\right)$

Example: coordinate ascent updates for a pairwise UGM

Let's consider applying this strategy to $p(x) \propto \exp\left(\sum_{i \sim j} \phi_{ij}(x_i, x_j)\right)$

Then, $E(x) = \sum_{i \sim j} \phi_{ij}(x_i, x_j)$. Let's assume $x_i \in \{+1, -1\}$.

The updates involve setting $q_i(X_i = x_i) := \frac{\exp\left(\mathbb{E}_{q_{-i}(x_{-i})} E(x)\right)}{Z_i}$

We can calculate, for $x_i \in \{1, -1\}$, the value of $\mathbb{E}_{q_{-i}(x_{-i})} E(x)$

Hence, $q_i(X_i = x_i) \propto \exp\left(\mathbb{E}_{q_{i,N(i)}(x_{i,N(i)})}\left(\sum_{j \in N(i)} \phi_{ij}(x_i, x_j)\right)\right)$

Note: the partition function Z_i only involves calculating the expression for $x_i = 1, -1$!

Example: coordinate ascent updates for a pairwise UGM

Let's consider applying this strategy to $p(x) \propto \exp\left(\sum_{i \sim j} \phi_{ij}(x_i, x_j)\right)$

Then, $E(x) = \sum_{i \sim j} \phi_{ij}(x_i, x_j)$. Let's assume $x_i \in \{+1, -1\}$.

The updates involve setting $q_i(X_i = x_i) := \frac{\exp\left(\mathbb{E}_{q_{-i}(x_{-i})} E(x)\right)}{Z_i}$

We can calculate, for $x_i \in \{1, -1\}$, the value of $\mathbb{E}_{q_{-i}(x_{-i})} E(x)$

Hence, $q_i(X_i = x_i) \propto \exp\left(\mathbb{E}_{q_{i,N(i)}(x_{i,N(i)})}\left(\sum_{j \in N(i)} \phi_{ij}(x_i, x_j)\right)\right)$

Somewhat reminiscent of Gibbs sampling, but deterministic.
*(You set it to the **expectation** of some quantity)*

The computer science approach: outer relaxation

Gibbs variational principle: Let $p(x) = \frac{1}{Z}\exp\big(E(x)\big)$ be a distribution over a domain $\mathcal{X}$. Then, Z is the solution to the following optimization problem:

$$\log Z = \max_{q\text{: distribution over } \mathcal{X}} H(q) + \mathbb{E}_{x\sim q}[E(x)]$$

What can we do to try to approximate this expression?

Inspiration from computer science: solve a *simpler* optimization problem over a *larger set* of mathematical objects that's easier to maximize over.

For example, often when we want to solve an integer program where the variables are required to be {0,1}, we instead solve a relaxed *linear program*, where the variables are required to be in [0,1].

The computer science approach: outer relaxation

Gibbs variational principle: Let $p(x) = \frac{1}{Z}\exp\big(E(x)\big)$ be a distribution over a domain $\mathcal{X}$. Then, Z is the solution to the following optimization problem:

$$\log Z = \max_{q:\ \text{distribution over } \mathcal{X}} H(q) + \mathbb{E}_{x\sim q}[E(x)]$$

Let's focus on UGMs: $p(x) \propto \exp\left(\sum_C \phi_C(x_C)\right)$

We can rewrite: $\mathbb{E}_q\left[\sum_C \phi_C(x_C)\right] = \sum_C \mathbb{E}_{q_C}[\phi_C(x_C)] = \sum_C \sum_{x_C} q_C(x_C)\,\phi_C(x_C)$

Objective can be rewritten as: $\max_{\{q_C(x_C)\}_C}\left\{\sum_C \mathbb{E}_{q_C}[\phi_C(x_C)] + \max_{\tilde{q}:\widetilde{q_C}(x_C)=q_C(x_C),\forall C} H(\tilde{q})\right\}$

The computer science approach: outer relaxation

Objective can be rewritten as: $\max_{\{q_C(x_C)\}_C} \left\{ \sum_C \mathbb{E}_{q_C}[\phi_C(x_C)] + \max_{\tilde{q}:\widetilde{q_C}(x_C)=q_C(x_C),\forall C} H(\tilde{q}) \right\}$

We seemingly gained something: we are now only optimizing over the set of all q_C. If there are K maximal cliques, each of size $\leq M$, and each variable x_i can take L values, we only need $K\ L^M$ variables to describe q_C. (As opposed to L^d.)

We have a different problem though: the set $\{q_C(x_C)\}_C$ needs to be such that there exists a distribution $\tilde{q}$, s.t. $\widetilde{q_C}(x_C) = q_C(x_C)$. In other words, these q_C satisfy some (implicit) constraints.

What are these constraints?

The computer science approach: outer relaxation

Objective can be rewritten as: $\max_{\{q_C(x_C)\}_C} \left\{ \sum_C \mathbb{E}_{q_C}[\phi_C(x_C)] + \max_{\tilde{q}:\widetilde{q_C}(x_C)=q_C(x_C),\forall C} H(\tilde{q}) \right\}$

Let's consider pairwise potentials, s.t. $|C| \leq 2$. Some obvious constraints:

$$\sum_{x_i} q_i(x_i) = 1, \qquad q_i(x_i) \geq 0, \forall x_i$$

$$\sum_{x_i} q_{i,j}(x_i, x_j) = q_j(x_j), \qquad q_{i,j}(x_i, x_j) \geq 0, \forall x_i, x_j$$

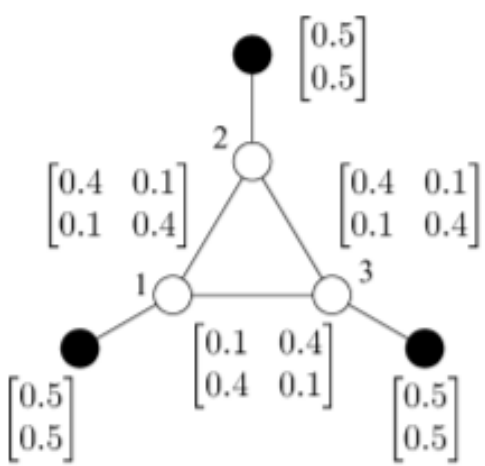
The polytope described by these (in)equalities is called the **local polytope**.

Is this all of them?

Must there be a distribution $\tilde{q}$, s.t. $\tilde{q}_{ij} = q_{ij}, \tilde{q}_i = q_i$?

The computer science approach: outer relaxation

Objective can be rewritten as: $$\max_{\{q_C(x_C)\}_C} \left\{ \sum_C \mathbb{E}_{q_C}[\phi_C(x_C)] + \max_{\tilde{q}:\widetilde{q_C}(x_C)=q_C(x_C),\forall C} H(\tilde{q}) \right\}$$



Example:

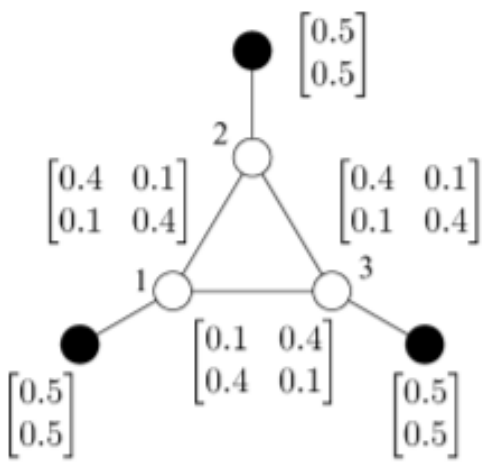
Consider an Ising model on a triangle graph as shown on the left.

Easy to check all the constraints on the previous slide are satisfied.

Fishy: *1 and 2, 2 and 3 have strong tendency to be the same; 1 and 3 do not. Can this* ***actually*** *happen?*

The computer science approach: outer relaxation

Objective can be rewritten as: $\max_{\{q_C(x_C)\}_C} \left\{ \sum_C \mathbb{E}_{q_C}[\phi_C(x_C)] + \max_{\tilde{q}:\widetilde{q_C}(x_C)=q_C(x_C),\forall C} H(\tilde{q}) \right\}$



Not quite. Notice:

$$1(x_1 \neq x_2) + 1(x_2 \neq x_3) + 1(x_1 = x_3) \geq 1$$

If $x_1 \neq x_3$, cannot be the case that both $x_1 = x_2$ and $x_2 = x_3$.

Taking expectations wrt $\tilde{q}$:

$$\mathrm{P}_{\tilde{q}}(x_1 \neq x_2) + \mathrm{P}_{\tilde{q}}(x_2 \neq x_3) + \mathrm{P}_{\tilde{q}}(x_1 = x_3) \geq 1$$

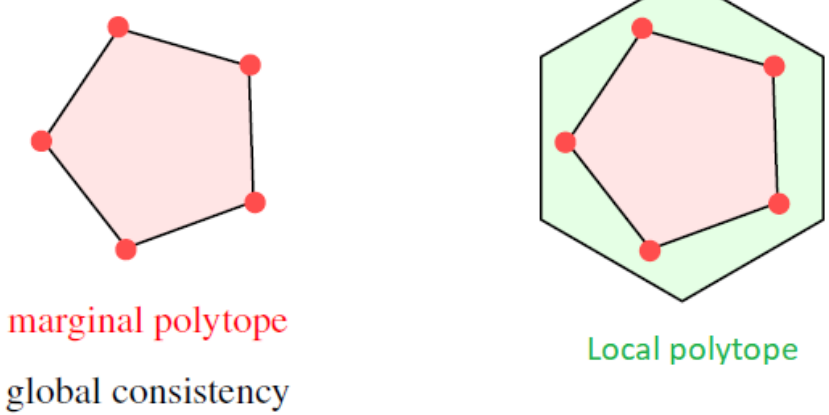
But, for distribution on picture: LHS = $0.2 + 0.2 + 0.2 < 1$

The computer science approach: outer relaxation

Objective can be rewritten as: $\max_{\{q_C(x_C)\}_C} \left\{ \sum_C \mathbb{E}_{q_C}[\phi_C(x_C)] + \max_{\tilde{q}:\widetilde{q_C}(x_C)=q_C(x_C),\forall C} H(\tilde{q}) \right\}$

The local polytope constitutes an "*outer approximation*" of the marginal polytope. (The polytope of q_C that are valid marginals of some distribution $\tilde{q}$.

This approach yields an approximation for log Z that is *more* than the true value (since we are maximizing over a bigger set than the original problem).



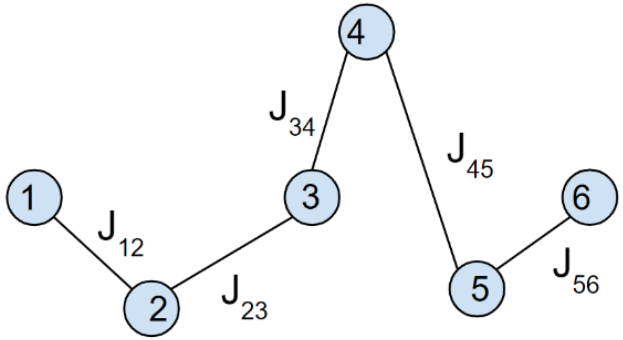
The computer science approach: outer relaxation

Objective can be rewritten as: $\max_{\{q_C(x_C)\}_C} \left\{ \sum_C \mathbb{E}_{q_C}[\phi_C(x_C)] + \max_{\tilde{q}:\widetilde{q_C}(x_C)=q_C(x_C),\forall C} H(\tilde{q}) \right\}$

In fact, this problem is more severe: in general, deciding whether $\{q_C(x_C)\}_C$ are valid marginals is NP-hard. (In other words, checking membership in the marginal polytope is NP-hard.)

Sometimes, the local approximation can be very good.

In fact, if UGM is a tree: it is exact.



Dealing with the entropy term

Objective can be rewritten as: $\max_{\{q_C(x_C)\}_C} \left\{ \sum_C \mathbb{E}_{q_C}[\phi_C(x_C)] + \max_{\tilde{q}:\widetilde{q_C}(x_C)=q_C(x_C),\forall C} H(\tilde{q}) \right\}$

Before we turn to trees, how do we deal with the term $\max_{\tilde{q}:\widetilde{q_C}(x_C)=q_C(x_C),\forall C} H(\tilde{q})$?

As written, evaluating the term seems to require solving an optimization problem.

In fact, it seems to require optimizing over all distributions $\tilde{q}$.
This was the problem we started out with ….

With an eye towards trees: what is the entropy $H(\tilde{q})$ for a tree?

Dealing with the entropy term

<u>Claim</u>: For a distribution $\tilde{q}$ following an (undirected) tree, and any rooting of the tree, we have

$$\mathrm{H}(\tilde{q}) = H(\tilde{q}(x_{\mathrm{root}})) + \sum_i H(\tilde{q}(x_i \,|x_{\mathrm{parent}(i)})$$

Moreover, for **any** distribution $\tilde{q}$ (not necessarily coming from a tree),

$$\mathrm{H}(\tilde{q}) \leq H(\tilde{q}(x_{\mathrm{root}})) + \sum_i H(\tilde{q}(x_i \,|x_{\mathrm{parent}(i)})$$

For any distribution $\tilde{q}$, we have $\tilde{q}(x_1, \dots x_n) = \tilde{q}(x_1)\tilde{q}(x_2|x_1) \dots \tilde{q}(x_n|x_1, \dots x_{n-1})$

For a tree $\tilde{q}$, by the global Markov property, rooting it at any node, we have

$$\tilde{q}(x_1, \dots x_n) = \tilde{q}(x_{\mathrm{root}})\Pi_i \, \tilde{q}(x_i|x_{\mathrm{parent}(i)})$$

Recall *chain rule for entropy:* $H(X, Y) = H(X) + H(Y|X)$, where

$$H(Y|X) \coloneqq \sum_x p(X = x)H(Y \,|X = x)$$

Dealing with the entropy term

<u>Claim</u>: For a distribution $\tilde{q}$ following an (undirected) tree, and any rooting of the tree, we have

$$\mathrm{H}(\tilde{q}) = H(\tilde{q}(x_{\mathrm{root}})) + \sum_i H(\tilde{q}(x_i \mid x_{\mathrm{parent}(i)})$$

Moreover, for **any** distribution $\tilde{q}$ (not necessarily coming from a tree),

$$\mathrm{H}(\tilde{q}) \leq H(\tilde{q}(x_{\mathrm{root}})) + \sum_i H(\tilde{q}(x_i \mid x_{\mathrm{parent}(i)})$$

For any distribution $\tilde{q}$, we have $\tilde{q}(x_1, \dots x_n) = \tilde{q}(x_1)\tilde{q}(x_2|x_1) \dots \tilde{q}(x_n|x_1, \dots x_{n-1})$

For a tree $\tilde{q}$, by the global Markov property, rooting it at any node, we have

$$\tilde{q}(x_1, \dots x_n) = \tilde{q}(x_{\mathrm{root}})\Pi_i\, \tilde{q}(x_i|x_{\mathrm{parent}(i)})$$

By the *chain rule for entropy*, $\mathrm{H}(\tilde{q}) = H(\tilde{q}(x_{\mathrm{root}})) + \sum_i H(\tilde{q}(x_i \mid x_{\mathrm{parent}(i)})$

Dealing with the entropy term

<u>Claim</u>: For a distribution $\tilde{q}$ following an (undirected) tree, and any rooting of the tree, we have

$$\mathrm{H}(\tilde{q}) = H(\tilde{q}(x_{\mathrm{root}})) + \sum_i H(\tilde{q}(x_i \,|x_{\mathrm{parent}(i)})$$

Moreover, for **any** distribution $\tilde{q}$ (not necessarily coming from a tree),

$$\mathrm{H}(\tilde{q}) \leq H(\tilde{q}(x_{\mathrm{root}})) + \sum_i H(\tilde{q}(x_i \,|x_{\mathrm{parent}(i)})$$

For any distribution $\tilde{q}$, we have $\tilde{q}(x_1, \dots x_n) = \tilde{q}(x_1)\tilde{q}(x_2|x_1) \dots \tilde{q}(x_n|x_1, \dots x_{n-1})$

Also by the chain rule for entropy, $\mathrm{H}(\tilde{q}) = H(\tilde{q}(x_1)) + \sum_i H(\tilde{q}(x_i|\, x_{<i}))$

Conditioning can only **decrease** entropy: $H(X|Y,Z) \leq H(X\,|Y)$

Hence, $\mathrm{H}(\tilde{q}) \leq H(\tilde{q}(x_{\mathrm{root}})) + \sum_i H(\tilde{q}(x_i \,|x_{\mathrm{parent}(i)})$ (take ordering $x_1, x_2, \dots, x_n$ to follow the ancestral ordering of the rooted tree).

Exactness on trees

We are optimizing:

$$\max_{\{q_{ij}(x_{ij})\}\in \boldsymbol{L}} \left\{ \sum_{i\sim j} \mathbb{E}_{q_{ij}}[\phi_{ij}(x_{ij})] + H(q(x_{\text{root}})) + \sum_i H(q(x_i \mid x_{\text{parent}(i)}) \right\} \ldots (1)$$

And trying to show it is equal to:

$$\max_{\{q_{ij}(x_{ij})\}\in \boldsymbol{M}} \left\{ \sum_{i\sim j} \mathbb{E}_{q_{ij}}[\phi_{ij}(x_{ij})] + \max_{\tilde{q}:\widetilde{q_{ij}}(x_C)=q_{ij}(x_C),\forall C} H(\tilde{q}) \right\} \ldots (2)$$

Since $\mathbf{M} \subseteq \boldsymbol{L}$ and *Claim*: $(1) \geq (2)$. How to prove $(1) \leq (2)$?

Strategy: Given the best solution to (1), produce a feasible solution to (2), that has the same value as (1). (Hence, $(1) \leq (2)$).

How to do this?

(1) We will produce a **distribution** q, whose pairwise marginals match q_{ij}

(2) Moreover, the entropy of q satisfies $H(q) = \max_{\tilde{q}:\widetilde{q_{ij}}(x_C)=q_{ij}(x_C),\forall C} H(\tilde{q})$

Exactness on trees

We are optimizing:

$$\max_{\{q_{ij}(x_{ij})\}\in \boldsymbol{L}} \left\{ \sum_{i\sim j} \mathbb{E}_{q_{ij}}[\phi_{ij}(x_{ij})] + H(q(x_{\text{root}})) + \sum_i H(q(x_i \mid x_{\text{parent}(i)}) \right\} \ldots (1)$$

And trying to show it is equal to:

$$\max_{\{q_{ij}(x_{ij})\}\in \boldsymbol{M}} \left\{ \sum_{i\sim j} \mathbb{E}_{q_{ij}}[\phi_{ij}(x_{ij})] + \max_{\tilde{q}:\widetilde{q_{ij}}(x_C)=q_{ij}(x_C),\forall C} H(\tilde{q}) \right\} \ldots (2)$$

Since $\mathbf{M} \subseteq \boldsymbol{L}$ and *Claim*: $(1) \geq (2)$. How to prove $(1) \leq (2)$?

Strategy: Given the best solution to (1), produce a feasible solution to (2), that has the same value as (1). (Hence, $(1) \leq (2)$).

How to do this? An algorithm to produce q:

Iterate through the nodes in the ancestral order.

For each node, produce sample for x_i by sampling from $q(x_i|x_{\text{parent}(i)})$

Exactness on trees

We are optimizing:

$$\max_{\{q_{ij}(x_{ij})\}\in \boldsymbol{L}} \left\{ \sum_{i\sim j} \mathbb{E}_{q_{ij}}[\phi_{ij}(x_{ij})] + H(q(x_{\text{root}})) + \sum_i H(q(x_i \mid x_{\text{parent}(i)}) \right\} \ldots (1)$$

And trying to show it is equal to:

$$\max_{\{q_{ij}(x_{ij})\}\in \boldsymbol{M}} \left\{ \sum_{i\sim j} \mathbb{E}_{q_{ij}}[\phi_{ij}(x_{ij})] + \max_{\tilde{q}:\widetilde{q_{ij}}(x_C)=q_{ij}(x_C),\forall C} H(\tilde{q}) \right\} \ldots (2)$$

How to do this? An algorithm to produce q:

Iterate through the nodes in the ancestral order.

For each node, produce sample for x_i by sampling from $q(x_i|x_{\text{parent}(i)})$

Marginals are preserved: For any two nodes $i\sim j$, $q(x_i, x_j) = q_{ij}(x_i, x_j)$

(Proof by induction)

Exactness on trees

We are optimizing:

$$\max_{\{q_{ij}(x_{ij})\}\in \boldsymbol{L}} \left\{ \sum_{i\sim j} \mathbb{E}_{q_{ij}}[\phi_{ij}(x_{ij})] + H(q(x_{\text{root}})) + \sum_i H(q(x_i \mid x_{\text{parent}(i)}) \right\} \ldots (1)$$

And trying to show it is equal to:

$$\max_{\{q_{ij}(x_{ij})\}\in \boldsymbol{M}} \left\{ \sum_{i\sim j} \mathbb{E}_{q_{ij}}[\phi_{ij}(x_{ij})] + \max_{\tilde{q}:\widetilde{q_{ij}}(x_C)=q_{ij}(x_C),\forall C} H(\tilde{q}) \right\} \ldots (2)$$

How to do this? An algorithm to produce q:

Iterate through the nodes in the ancestral order.

For each node, produce sample for x_i by sampling from $q(x_i|x_{\text{parent}(i)})$

Entropy is preserved: The entropy of the distribution q is (by *Claim*)

$$\mathrm{H}(q) = H(q(x_{\text{root}})) + \sum_i H\big(q(x_i \mid x_{\text{parent}(i)}\big)$$

Also by *Claim*: $\max_{\tilde{q}:\widetilde{q_{ij}}(x_C)=q_{ij}(x_C),\forall C} H(\tilde{q}) \leq H(q)$.

What if the graph is not a tree?

The expression for entropy we chose

$$\mathrm{H}(\tilde{q}) = H(\tilde{q}(x_{\mathrm{root}})) + \sum_i H(\tilde{q}(x_i \mid x_{\mathrm{parent}(i)})$$

does not even make sense: you cannot "root" a graph with loops.

How do we even approximate the term $\max_{\tilde{q}:\widetilde{q_{ij}}(x_C)=q_{ij}(x_C),\forall C} H(\tilde{q})$?

Rewriting the entropy

<u>Claim</u>: If the graph is a **tree,** the entropy $H(q)$ can be written as follows:

$$H(q) = \sum_{i \sim j} H(q_{ij}) - \sum_{i} (d_i - 1) H(q_i)$$

For a tree, we had $\mathrm{H}(\tilde{q}) = H(\tilde{q}(x_{\mathrm{root}})) + \sum_{i} H(\tilde{q}(x_i \,|x_{\mathrm{parent}(i)})$

By the chain rule for entropy, we have

$$H\big(\tilde{q}(x_i \,|x_{\mathrm{parent}(i)}\big) = H\big(\tilde{q}(x_i, x_{\mathrm{parent}(i)}\big) - H(\tilde{q}\big(x_{\mathrm{parent}(i)}\big))$$

Hence, every "edge" entropy is added once; every "node" entropy is added once, and subtracted d_i times.

Thus, the claim follows.

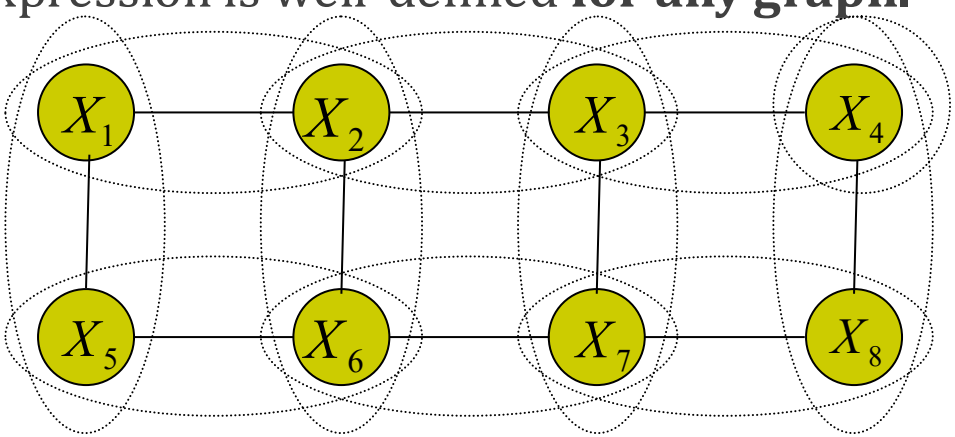
Rewriting the entropy

<u>Claim</u>: If the graph is a **tree,** the entropy $H(q)$ can be written as follows:

$$H(q) = \sum_{i \sim j} H(q_{ij}) - \sum_{i} (d_i - 1) H(q_i)$$

Interpretation: every "edge" entropy is added once. Each "node" is "counted" for every edge it participates in, so we should "subtract" the extra $(d_i - 1)$ times it's counted.

Benefit of rewrite: the above expression is well-defined **for any graph.** (Whether it is a tree or not.)



$$H_{\text{Bethe}}(q) = H(q_{12}) + H(q_{23}) + \ \ldots + H(q_{67}) + H(q_{78}) - H(q_1) - 2H(q_2) - 2H(q_6) \ldots$$

Rewriting the entropy

<u>Claim</u>: If the graph is a **tree,** the entropy $H(q)$ can be written as follows:

$$H(q) = \sum_{i \sim j} H(q_{ij}) - \sum_i (d_i - 1) H(q_i)$$

Interpretation: every "edge" entropy is added once. Each "node" is "counted" for every edge it participates in, so we should "subtract" the extra $(d_i - 1)$ times it's counted.

Benefit of rewrite: the above expression is well-defined **for any graph.** (Whether it is a tree or not.)

$H_{Bethe}(q) := \sum_{i \sim j} H(q_{ij}) - \sum_i (d_i - 1) H(q_i)$
is called the **Bethe entropy,** after the physicist Hans Bethe.

The Bethe relaxation

Regardless if the graph is a tree or not, we can write down the following relaxation to the variational principle for the log partition function:

$$\max_{\{q_{ij}(x_{ij})\}\in \boldsymbol{M}} \left\{ \sum_{i\sim j} \mathbb{E}_{q_{ij}}[\phi_{ij}(x_{ij})] + \; H_{\text{Bethe}}(q) \right\}$$

The above quantity is also called the Bethe free energy.

It is not guaranteed to be either a lower or upper bound of log Z:

$H_{\text{Bethe}}(q)$ can be both less and more than log Z.

It is generally not going to equal the entropy.

It is generally not going to be concave.

The Bethe relaxation

Regardless if the graph is a tree or not, we can write down the following relaxation to the variational principle for the log partition function:

$$\max_{\{q_{ij}(x_{ij})\}\in \boldsymbol{M}} \left\{ \sum_{i\sim j} \mathbb{E}_{q_{ij}}[\phi_{ij}(x_{ij})] + \; H_{\text{Bethe}}(q) \right\}$$

We can rewrite the Bethe entropy in a slightly more convenient form:

$$H_{\text{Bethe}}(q) = -\sum_{i\sim j}\sum_{x_i,x_j} q_{ij}(x_i,x_j)\log\frac{q_{ij}(x_i,x_j)}{q_i(x_i)q_j(x_j)} - \sum_{i}\sum_{x_i} q_i(x_i)\log q_i(x_i)$$

Can we try to solve it despite non-concavity?

Since this is a constrained maximization problem, one reasonable way to solve it is via introducing Lagrange multipliers:

$$\max_{q_i(x_i), q_{ij}(x_{ij}), \lambda_i, \lambda_{i,j}(x_j)} \sum_{i \sim j} \mathbb{E}_{q_{ij}}\left[\phi_{ij}(x_{ij})\right] + H_{\text{Bethe}}(q) + \sum_{i=1}^{d} \lambda_i \left(\sum_{x_i} q_{i(x_i)} - 1 \right)$$

$$+ \sum_{i \sim j} \left(\sum_{x_j} \lambda_{ij}(x_j) \left(\sum_{x_i} q_{ij(x_i, x_j)} - q_{j(x_j)} \right) + \sum_{x_i} \lambda_{ji}(x_i) \left(\sum_{x_j} q_{ij(x_i, x_j)} - q_{i(x_i)} \right) \right)$$

What are the stationary points of this objective? (i.e. points where grad = 0)

$$0 = \frac{\partial L}{\partial q_{ij}(x_i, x_j)} = \phi_{ij}(x_{ij}) - \log q_{ij}(x_i, x_j) + \log\left(q_i(x_i) q_j(x_j)\right) + \left(\lambda_{ij}(x_j) + \lambda_{ji}(x_i)\right) + \text{const}$$

$$\Rightarrow q_{ij}(x_i, x_j) \propto q_i(x_i) q_j(x_j) \exp\left(\phi_{ij}(x_{ij}) + \left(\lambda_{ij}(x_j) + \lambda_{ji}(x_i)\right)\right)$$

Can we try to solve it despite non-concavity?

Since this is a constrained maximization problem, one reasonable way to solve it is via introducing Lagrange multipliers:

$$\max_{q_i(x_i), q_{ij}(x_{ij}), \lambda_i, \lambda_{i,j}(x_j)} \sum_{i \sim j} \mathbb{E}_{q_{ij}}\left[\phi_{ij}(x_{ij})\right] + \; H_{\text{Bethe}}(q) + \sum_{i=1}^{d} \lambda_i \left(\sum_{x_i} q_{i(x_i)} - 1 \right)$$

$$+ \sum_{i \sim j} \left(\sum_{x_j} \lambda_{ij}(x_j) \left(\sum_{x_i} q_{ij(x_i, x_j)} - q_{j(x_j)} \right) + \sum_{x_i} \lambda_{ji}(x_i) \left(\sum_{x_j} q_{ij(x_i, x_j)} - q_{i(x_i)} \right) \right)$$

What are the stationary points of this objective? (i.e. points where grad = 0)

$$0 = \frac{\partial L}{\partial q_i(x_i)} = \; -\log q_i(x_i) + \sum_{i \sim j} \sum_{x_j} \frac{q_{ij}(x_i, x_j)}{q_i(x_i)} + \lambda_i - \sum_j \lambda_{ji}(x_i) + \text{const}$$

$$\Rightarrow q_i(x_i) \propto \exp\left(-\sum_{i \sim j} \lambda_{ji}(x_i) \right)$$

Can we try to solve it despite non-concavity?

$$q_{ij}(x_i, x_j) \propto q_i(x_i) q_j(x_j) \exp\left(\phi_{ij}(x_{ij}) + \left(\lambda_{ij}(x_j) + \lambda_{ji}(x_i)\right)\right)$$

$$q_i(x_i) \propto \exp\left(-\sum_{i \sim j} \lambda_{ji}(x_i)\right)$$

Additionally, we have $\sum_{x_j} \; q_{ij}(x_i, x_j) = q_i(x_i)$:

$$q_i(x_i) \propto q_i(x_i) \exp\left(\lambda_{ji}(x_i)\right) \sum_{x_j} q_j(x_j) \exp\left(\phi_{ij}(x_{ij}) + \lambda_{ij}(x_j)\right)$$

$$\exp\left(-\lambda_{ji}(x_i)\right) \propto \sum_{x_j} q_j(x_j) \exp\left(\phi_{ij}(x_{ij}) + \lambda_{ij}(x_j)\right)$$

$$\exp\left(-\lambda_{ji}(x_i)\right) \propto \sum_{x_j} \exp\left(-\sum_{k \sim j} \lambda_{kj}(x_j) + \phi_{ij}(x_{ij}) + \lambda_{ij}(x_j)\right)$$

Can we try to solve it despite non-concavity?

$$q_{ij}(x_i, x_j) \propto q_i(x_i) q_j(x_j) \exp\left(\phi_{ij}(x_{ij}) + \left(\lambda_{ij}(x_j) + \lambda_{ji}(x_i)\right)\right)$$

$$q_i(x_i) \propto \exp\left(-\sum_{i \sim j} \lambda_{ji}(x_i)\right)$$

Additionally, we have $\sum_{x_j} \; q_{ij}(x_i, x_j) = q_i(x_i)$:

$$q_i(x_i) \propto q_i(x_i) \exp\left(\lambda_{ji}(x_i)\right) \sum_{x_j} q_j(x_j) \exp\left(\phi_{ij}(x_{ij}) + \lambda_{ij}(x_j)\right)$$

$$\exp\left(-\lambda_{ji}(x_i)\right) \propto \sum_{x_j} q_j(x_j) \exp\left(\phi_{ij}(x_{ij}) + \lambda_{ij}(x_j)\right)$$

$$\exp\left(-\lambda_{ji}(x_i)\right) \propto \sum_{x_j} \exp\left(-\sum_{k \sim j, k \neq i} \lambda_{kj}(x_j) + \phi_{ij}(x_{ij})\right)$$

Can we try to solve it despite non-concavity?

$$q_{ij}(x_i, x_j) \propto q_i(x_i) q_j(x_j) \exp\left(\phi_{ij}(x_{ij}) + \left(\lambda_{ij}(x_j) + \lambda_{ji}(x_i)\right)\right)$$

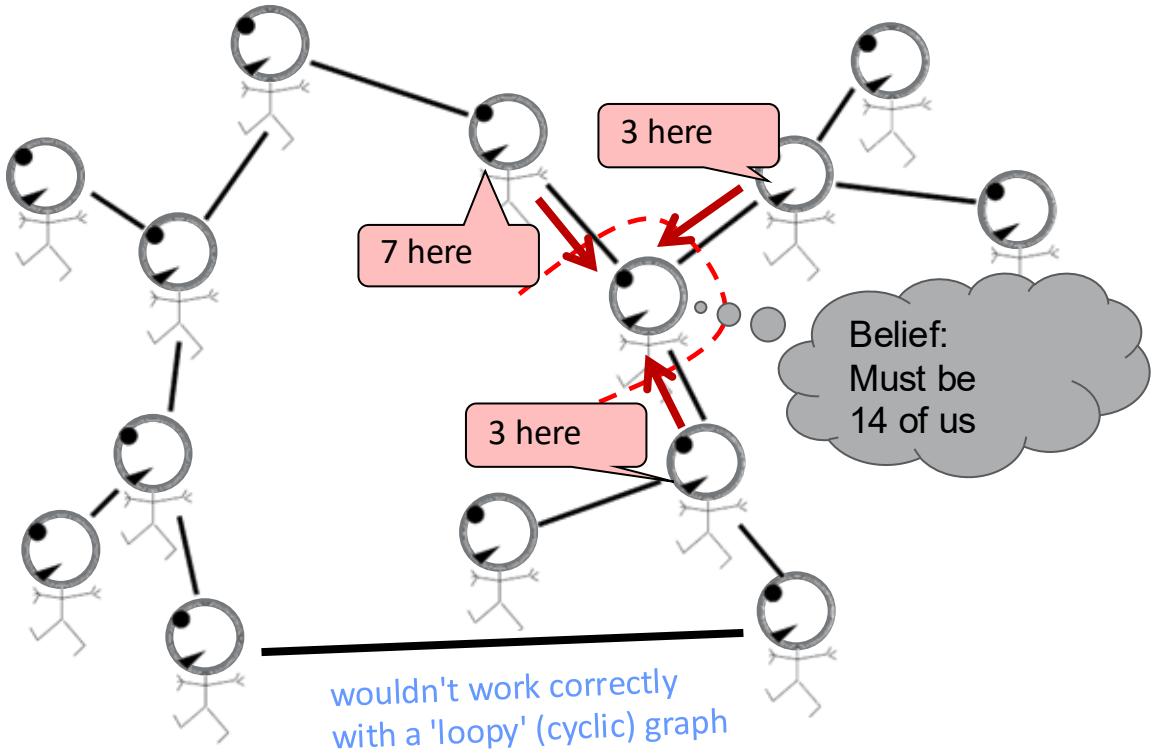
$$q_i(x_i) \propto \exp\left(-\sum_{i \sim j} \lambda_{ji}(x_i)\right)$$

$$\exp\left(-\lambda_{ji}(x_i)\right) \propto \sum_{x_j} \exp\left(-\sum_{k \sim j, k \neq i} \lambda_{kj}(x_j) + \phi_{ij}(x_{ij})\right)$$

Potential idea: keep updating the LHS by the value of the RHS, until you (hopefully?) reach a fixed point.

Remind you of something?

Recall: Message Passing



Carnegie
Mellon
University

Can we try to solve it despite non-concavity?

$$q_{ij}(x_i, x_j) \propto q_i(x_i) q_j(x_j) \exp\Big(\phi_{ij}(x_{ij}) + \big(\lambda_{ij}(x_j) + \lambda_{ji}(x_i)\big)\Big)$$

$$q_i(x_i) \propto \exp\left(-\sum_{i\sim j} \lambda_{ji}(x_i)\right)$$

$$\exp\big(-\lambda_{ji}(x_i)\big) \propto \sum_{x_j} \exp\left(-\sum_{k\sim j, k\neq i} \lambda_{kj}(x_j) + \phi_{ij}(x_{ij})\right)$$

Remember message passing?

$$m_{j\to i}(x_i) = \sum_{x_j} \exp(\phi_{ij}(x_i, x_j)) \prod_{k\in N(j)\setminus i} m_{k\to j}(x_j).$$

Set $\mathrm{m}_{j\to i}(x_i) = \exp\big(-\lambda_{ji}(x_i)\big)$

Can we try to solve it despite non-concavity?

$$\exp\left(-\lambda_{ji}(x_i)\right) \propto \sum_{x_j} \exp\left(-\sum_{k\sim j, k\neq i} \lambda_{kj}(x_j) + \phi_{ij}(x_{ij})\right)$$

$$m_{j\to i}(x_i) = \sum_{x_j} \exp(\phi_{ij}(x_i, x_j)) \prod_{k\in N(j)\setminus i} m_{k\to j}(x_j).$$

Set $\mathrm{m}_{j\to i}(x_i) = \exp\left(-\lambda_{ji}(x_i)\right)$

Formal connection between running belief propagation on graphs with loops and Bethe relaxation:

Stationary point of the Lagrangian formulation of Bethe relaxation are in **one-to-one correspondence** with fixed points of the belief propagation updates.

(By mapping $\mathrm{m}_{j\to i}(x_i)$ to $\exp\left(-\lambda_{ji}(x_i)\right)$)

Gives us license to run loopy belief propagation: it would correspond to trying to find stationary points of Bethe relaxation.

Can we try to solve it despite non-concavity?

$$\exp\left(-\lambda_{ji}(x_i)\right) \propto \sum_{x_j} \exp\left(-\sum_{k\sim j, k\neq i} \lambda_{kj}(x_j) + \phi_{ij}(x_{ij})\right)$$

$$m_{j\to i}(x_i) = \sum_{x_j} \exp(\phi_{ij}(x_i, x_j)) \prod_{k\in N(j)\setminus i} m_{k\to j}(x_j).$$

Set $\mathrm{m}_{j\to i}(x_i) = \exp\left(-\lambda_{ji}(x_i)\right)$

Caveats:

May or may not converge. (No formal characterization when.)

Even if it converges, it only does so to a stationary point. (May or may not be close to the global optimum, no formal characterization when.)

Some empirical studies on a few famous graphical models like QMR-DT:

[Loopy-belief Propagation for Approximate Inference: An Empirical Study]
Kevin Murphy, Yair Weiss, and Michael Jordan.
UAI '99 (Uncertainty in AI). **]**

A short glimpse of relaxations

We saw the most "basic" relaxations. Lots of further work over the years.

Definitive reference:

Graphical Models, Exponential Families, and Variational Inference

Martin J. Wainwright[1] and Michael I. Jordan[2]

There's approaches to "convexify" Bethe entropy (e.g. tree-reweighted BP, sum-of-squares and convex programming hierarchies, etc.) .

There's approaches to "increase" the locality (e.g. Kikuchi approximations).

There's approaches to "correct" the Bethe entropy (e.g. loop series expansions; usually work when there's not too many / short loops).